SUMANTH R

Chennai, Tamil Nadu
B.Tech. in Computer Science and Engineering (AIML)
SRM Institute of Science and Technology

+91-9344249005
rsumanth2005@gmail.com
LinkedIn
GitHub

Education

Degree	Institute	CGPA	Duration
B.Tech., Computer Science and Engineering (AIML)	SRM Institute of Science and Technology, Chennai, Tamil Nadu	8.98 / 10	*2023 – 2027*

Experience

- **Works Applications India Pvt. Ltd.** *Dec 2025 – Jan 2026*
  Software Engineering Intern On-site
  - Built a full-stack application using **Java, Spring Boot, MySQL, and React**, developing 5+ REST APIs for business workflows.
  - Optimized SQL queries and database indexing, reducing backend response time by **25%** during testing.
  - Integrated **Explainable AI** insights into the frontend to make model outputs more interpretable and actionable.

- **All India Council for Technical Education (AICTE)** *July 2024 – September 2024*
  Google AI/ML Virtual Intern Remote
  - Built and evaluated **8+ ML models** across supervised and unsupervised learning tasks on 5+ datasets.
  - Applied **data preprocessing, feature engineering, and hyperparameter tuning** to improve model performance.
  - Studied real-world AI use cases and applied **bias mitigation** techniques to improve model reliability and fairness.

Projects

- **Market Intelligence & Competitive Benchmarking** *2026*
  Technologies: Python, SQL, Pandas, Streamlit, Plotly, REST APIs
  - Built a market intelligence workflow to analyze **20+ business metrics** across companies, markets, and competitors.
  - Developed analytical models to evaluate **growth, profitability, market share, and competitive positioning**, identifying key performance drivers.
  - Automated data processing and interactive dashboards to convert fragmented data into **actionable insights and executive-ready visualizations**.

- **Customer Segmentation & Pricing Analytics** *2026*
  Technologies: Python, SQL, Pandas, Scikit-learn, Matplotlib
  - Analyzed customer transaction data using **RFM analysis and clustering** to identify high-value, growth, and at-risk customer segments.
  - Evaluated **customer behavior, pricing sensitivity, and revenue potential** across segments using statistical and scenario-based analysis.
  - Translated analytical findings into **data-backed pricing and retention recommendations** to support business decision-making.

- **AI-Assisted Industry Research & Company Screening** *2026*
  Technologies: Python, LLMs, RAG, SQL, LangChain, Streamlit
  - Built an AI-assisted research workflow to extract, structure, and compare information from **company reports, industry documents, and datasets**.
  - Implemented **RAG-based retrieval and structured scoring** to evaluate companies across market, financial, operational, and competitive dimensions.
  - Generated standardized company profiles and research summaries to accelerate **hypothesis generation, comparative analysis, and decision-making**.

Publications

- **Deep Learning Based Brain Tumor Classification using Orange Data Analytics** *2026*
  IEEE International Conference (SCEECS 2026) View Paper
  - Analyzed a **16,000+ MRI image dataset** aggregated from Kaggle, BRISC, and FigShare to develop a robust classification pipeline.
  - Applied **PCA, data augmentation, and stratified 10-fold cross-validation**, reducing feature dimensionality by **75%** while retaining **95% variance**.
  - Achieved **96%+ classification accuracy** and **0.997 AUC**, improving performance by **7% over baseline methods** and MCC by **8%**.

Achievements

- Selected as a **Software Engineering Fellow at Headstarter AI**, working on real-world software engineering and AI projects.
- Secured **3rd Place** in the **CodeMavens 2025 Hackathon** organized by the ACM-W SRM Student Chapter.
- Published research on **deep learning-based brain tumor classification** at the **IEEE International Conference (SCEECS 2026)**.

Technical Skills

- **Programming**: Python, SQL, Java, C++
- **Data Analysis**: Pandas, NumPy, Scikit-learn, Statistical Analysis, Data Cleaning, Exploratory Data Analysis
- **Analytics & Visualization**: Data Visualization, Matplotlib, Plotly, Streamlit, Dashboard Development
- **AI / Generative AI**: Machine Learning, NLP, LLMs, RAG, LangChain, Hugging Face
- **Databases & Tools**: MySQL, MongoDB, Git, GitHub, Jupyter Notebook, Linux
- **Core Concepts**: Data Structures & Algorithms, OOP, DBMS, Analytical Problem Solving, Hypothesis-Driven Analysis